

UFFT: Adversarial Referee Audit

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6 October 2026

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Recommendation

Do not accept the present claim that UFFT derives the Standard Model and general relativity from one axiom with no fitted parameters. The public derivation contains explicit algebraic errors, assumes important parts of the proposed outcome, and has inconsistent status declarations across papers, scripts and website. This conclusion is about the supplied arguments, not a proof that every possible foam theory is impossible.

A substantially narrower paper on the face graph, its representation decomposition and carefully delimited spectral characterisations could merit mathematical review. A physical programme would need a specified microscopic action and a new derivation of its interacting continuum limit. Correcting the identified arithmetic and coding errors alone would not establish that limit.

Scope and evidence standard

This is an initial referee audit with corpus-wide screening and targeted detailed checks, not a completed line-by-line proof certification of every paper. “ReviewBot” denotes the review format; this report was produced by one assistant, not independent human referees or multiple independently working agents.

The frozen public repository snapshot is commit `78f9a289842e6de49ecbc886d1bbca19f5e7f925`, retrieved during this audit. The repository contains 85 paper Markdown files, excluding `INDEX.md`. The index has 81 rows linking paper Markdown files to Zenodo records, including repeated versions, standalone and unnumbered entries. Its headline count is not an independently established count of current unique papers. Two indexed alpha versions (#3 v2 and v3) are absent as Markdown files in this snapshot. No numbered #4 or #12 is present in the index. The public scope therefore cannot safely be described as “77 independently audited papers.”

The paper-screening appendix covers every indexed row and lists unindexed repository files. Screening uses the indexed principal claims, available abstracts, status statements and selected argument passages. Detailed mathematical/code checks concentrate on #5, #48, #50, #57–60, #63, #68, #70, #72–76, the standalone Fedorov note, and the numerical-selection machinery. Other rows explicitly remain screening judgments rather than full derivation audits. Zenodo record access failed through the web tool for selected records; the author’s repository versions were used, without asserting byte identity with archived PDFs or DOCX files.

Eight of nine initially selected author scripts completed. The original spectrum script could not start because sympy is missing in this environment; this is an environment limitation, not evidence against its mathematics. A tenth script, the hydrogen unit map, subsequently completed. An independent NumPy-only check reconstructs the graph and computes its characteristic polynomial exactly using integer traces and Newton identities. All logs and the independent check are attached. A zero process exit is recorded as successful execution, not scientific endorsement; some scripts print failing evidence gates or unproved interpretations while exiting zero.

1. What is confirmed

The face-adjacency graph has 14 vertices and 36 edges. Building it directly from six axial square-face normals and eight sign-vector hexagonal normals reproduces

$$p_L(x) = x(x-9)(x-7)^4(x-4)^2(x^2-9x+16)^3.$$

The graph has 24 triangles and 42 simple four-cycles. The quadratic discriminant is 17. The author’s character calculation reproduces

$$\rho = 2A_{1g} \oplus E_g \oplus 2T_{1u} \oplus T_{2g} \oplus A_{2u}.$$

The inter-orbit block has the stated Pauli form in the chosen multiplicity basis. The rank-12 projector obtained by removing the eigenvalue-0 and eigenvalue-9 modes has trace fraction 6/7. These are real combinatorial/algebraic results. Their identification with particles, chirality, generations, vacuum energy or gauge interactions is a separate claim.

The five-cell comparison script reproduces the advertised spectra for its supplied graph constructions. That supports the restricted spectral characterisation, but a journal proof should also justify every graph construction against the relevant polytope. The script includes two alternative elongated-dodecahedron constructions and is not by itself a geometric classification proof. Originality and priority of the graph results have not been established by this audit.

2. Blocking defects in the central physical argument

R1. Gauge groups are supplied to the derivation

Paper #59 §6.4(ii) specifies inter-cell links taking values in the full $SU(2)$ group, and §2 introduces the ordinary gauge-link transformation and a Wilson plaquette action. Obtaining $SU(2)$ Yang–Mills from $SU(2)$ links does not derive $SU(2)$ from a finite face graph. The analogous $SU(3)$ claim has the same dependency.

The stated starting quadratic face action also lacks an explicit temporal discretisation, full inter-cell coupling prescription, fermionic measure, independent gauge degrees of freedom and gravitational variables. Plaquette terms, Yukawa terms and a quartic potential are added in the argument rather than derived from a documented integration over the original degrees of freedom. A quadratic action with fixed coefficients describes a Gaussian theory; nontrivial emergent interactions require a specified mechanism.

Required correction: state all microscopic variables and the full action, distinguish fixed phases from dynamical group-valued links, and prove the effective group and interactions without selecting them in advance.

R2. The displayed Dirac construction does not supply a 3+1-dimensional Clifford algebra

Paper #59 §3.3 writes gamma-zero as sigma-z, gamma-i as sigma-x times a spatial derivative, and gamma-five as sigma-z. A gamma matrix is not a differential operator. Reusing sigma-x for different spatial directions fails the required anticommutation between those directions. Taking gamma-five equal to gamma-zero fails their anticommutation as well. Extra spinor indices and correct matrices might repair a model, but they are not supplied by this displayed argument.

Paper #72's positive square root of a positive graph Laplacian is valid spectral mathematics. It does not by itself establish locality, Lorentz spin, fermionic statistics, a chiral gauge representation or the claimed doubler removal. Moreover, its discussion of positive and negative square roots being related by a chirality involution cannot hold as unitary conjugation of a nonzero positive operator: conjugation preserves its spectrum.

Required correction: construct an actual lattice fermion operator, with spinor space, Clifford relations, locality, gauge coupling and a consistent treatment of doublers.

R3. The proposed Yukawa expression is identically zero

Paper #59 §4.2 evaluates a matrix element containing

$$TP_{A_{2u}}T$$

as proportional to minus-four times the identity. Yet corrected Paper #57 explicitly states that the inter-type operator T annihilates A_{2u} . In the specified graph, T is real antisymmetric and the Higgs vector is in its kernel. Consequently both $T P_{A_{2u}}$ and $P_{A_{2u}} T$ vanish. Our direct reconstruction gives

$$\|TP_{A_{2u}}T\| = 0.$$

Replacing a projector-mediated coupling with the scalar charge of a different operator does not repair this identity. A new interaction tensor could be proposed, but would need a separate definition and derivation. There is also a representation selection issue: two odd T_{1u} fields and an odd A_{2u} field cannot form an even invariant under the full stated O_h action by their ordinary tensor product alone.

Verdict: the displayed Yukawa derivation fails, not merely lacks experimental confirmation.

R4. The Higgs mass sign contradicts the printed formula

Paper #59 §5.1 prints a mass-squared proportional to $-(\text{lambda_A2u} * \text{tau_A2u})$ and calls it negative. Substituting $\text{lambda}=9$ and $\text{tau}=-1$ gives $+9$ times the positive scale factor. That potential has a stable origin for positive quartic coupling, contrary to the stated symmetry-breaking argument.

Even after a sign correction, a negative adjacency eigenvalue does not automatically imply a negative physical mass-squared. Paper #57 introduces that sign mapping as a physical rule. Also, a one-dimensional spatial irrep does not supply the four real components of a complex electroweak Higgs doublet or its three Goldstone directions. Those require additional internal degrees of freedom.

Required correction: derive the potential and its internal representation from the microscopic action, rather than infer it from adjacency sign.

R5. The anomaly closure contains an explicit contradictory limit

Paper #60 §2.1 defines $\Gamma_5 = T/(2i)$ and uses $T\text{-squared}=-4I$. This gives exactly

$$\{T, \Gamma_5\} = 4iI.$$

The next passage notes that an expression proportional to lattice spacing tends to zero and concludes that this same anticommutator tends to zero. With the displayed fixed operators, it does not. Its independently computed Frobenius norm on the six-dimensional T_{1u} sector is $4\sqrt{6}$, approximately 9.798. No consistent spacing-dependent operator family is supplied to justify the claimed limit.

A Ginsparg–Wilson relation also does not automatically imply absence of all doublers without appropriate further conditions. The anomaly discussion conflates axial anomalies with gauge anomaly cancellation and does not enumerate the complete chiral hypercharge representations. A Laplacian zero mode is not an anomaly-cancellation calculation.

Verdict: Theorem 60.1 is not proved by the supplied calculation.

R6. The continuum verification uses a non-Hermitian matrix as though it were Hermitian

In `Symanzik_Matching_BCC.py`, the Bloch matrix diagonal receives $1 - \exp(i \mathbf{k} \cdot \boldsymbol{\delta})$ for each face, without its reverse-hop conjugate on that same diagonal. Generic diagonal entries therefore have nonzero imaginary parts. At $\mathbf{k}=(0.2,0.3,0.4)$, our call to the supplied builder gives

$$\|H - H^\dagger\|_F \approx 2.10463,$$

and maximum imaginary diagonal magnitude approximately 0.43497. The script passes H to `numpy.linalg.eigvalsh`, which assumes Hermiticity and treats the diagonal as real. It is not validating the spectrum of the matrix it has constructed. Several final matching coefficients are also stated or hard-coded rather than extracted as a complete interacting effective theory.

This is a defect in this supplied script. It does not imply that every Bloch model elsewhere in the repository is non-Hermitian. It does mean this script cannot support the claimed closure of #59–60.

Cubic symmetry may constrain the quadratic scalar dispersion to be isotropic, but it does not by itself impose Lorentz symmetry, a common limiting speed in all sectors, absence of all marginal anisotropies, or diffeomorphism invariance. Asymptotic freedom of a selected gauge sector is not a proof of uniqueness of the complete SM+GR continuum theory.

Required correction: derive a Hermitian full lattice operator, verify reverse-hop consistency, state the effective operator basis and matching prescription, and check all relevant and marginal terms.

R7. The alleged finite-group embedding is wrong as stated

Papers #58–59 describe D_3 , isomorphic to S_3 , as a subgroup of $SU(2)$, appealing to its binary dihedral lift. These are different groups. $SU(2)$ has only one nonidentity order-two element, minus- I . S_3 has three. Hence S_3 cannot embed faithfully as a subgroup of $SU(2)$. The binary dihedral group can embed, with S_3 obtained as a quotient.

This correction is elementary and leaves the physical emergence problem open. Spatial rotations becoming more isotropic also does not turn an internal finite-group action into a local $SU(2)$ gauge redundancy.

R8. The GR closure misuses both elasticity and Weinberg–Witten

Paper #60 §4.2 sets a metric perturbation $h_{ij} = \partial_i u_j + \partial_j u_i$. About flat space, that is a pure coordinate deformation. Extended consistently as a diffeomorphism, its linearised Riemann tensor vanishes. Elastic strain energy for this displacement does not establish a propagating gravitational curvature field or the Einstein–Hilbert action. A more general metric order parameter would have to be introduced and its constraints derived.

Section 4.3 presents Weinberg–Witten as a theorem guaranteeing that a Lorentz-covariant massless spin-two excitation yields nonlinear GR. Its actual result is an obstruction under specified conserved-current/stress-tensor assumptions. GR evades those assumptions; one must explain the appropriate evasion in an emergent model. Soft-graviton or self-coupling arguments are distinct results and need their own hypotheses. See Weinberg and Witten, *Limits on massless particles*, Phys. Lett. B 96 (1980), 59–62, DOI 10.1016/0370-2693(80)90212-9.

Removing the Eg part of the $SO(3)$ spin-two representation because it is assigned to another sector also does not leave a complete $SO(3)$ spin-two multiplet. Counting an Oh shear component

is not a proof of two massless graviton helicities.

Paper #60 §4.4 mixes reduced and unreduced Planck masses: its displayed $G = \hbar c/M_P^2$ is incompatible with its displayed $M_P = \sqrt{\hbar c/(8 \pi G)}$. The overall dimensional scale is an input, not obtained from pure cell integers.

Verdict: Theorem 60.3 does not establish GR.

3. Representation and ontology problems

The representation multiplicity of T1u in the 14-dimensional face space is **two**, while its irrep dimension is **three**. Paper #72 correctly lists multiplicity two but repeatedly calls the three-generation identification a multiplicity theorem. Threefold eigenvalue degeneracy is not three copies of the irrep. Identifying its three spatial components as flavor generations requires explaining why ordinary rotations do not mix physically distinct flavors and how distinct flavor masses arise compatibly with rotational invariance. Schur's lemma gives degeneracy for an invariant operator on this irrep, not the observed unequal masses.

Spatial inversion parity is not fermion statistics or chirality. Indeed A2u is odd and is identified as the bosonic Higgs. A triplet of spatial modes is not automatically the eight generators of the SU(3) adjoint. Spatial and internal representation indices need separate definitions.

The finite single-cell catalogue does not prove that multi-cell composites or collective excitations cannot add sectors. The corpus itself appeals to a collective graviton in #60; excluding all extra sectors by face-space exhaustion while allowing that collective sector is inconsistent without a separate completeness theorem. Thus no-SUSY and no-fourth-generation results are hypotheses of a chosen ontology, not consequences about Nature of rank-nullity alone.

The chirality/gauge assignment is not consistent across current sources: #57–58 use square-supported Eg for electroweak placement and band-based handedness, while #72 uses a conditional SU(2)L-adjoint-to-T2g embedding on the hex orbit. In #58, that same T2g is assigned to color. The authors must specify whether a tensor factor or other independent structure reconciles these meanings; the present text does not.

4. Geometry and restricted uniqueness

The site's global minimum-area Kelvin assertion is false as written: Weaire and Phelan supplied a lower-area equal-volume foam. See their 1994 *Philosophical Magazine Letters* paper, DOI 10.1080/09500839408241577. This does not invalidate the truncated-octahedron graph calculation.

Paper #50 v2 does narrow its spectral uniqueness claim to five combinatorial types of convex translational parallelotopes and excludes multi-cell foams axiomatically. That restriction and the preference for a prime discriminant are additional selection premises, not consequences established from $B+V=D$. Combinatorial spectral uniqueness is not a surface-area minimisation theorem; neither implies a microscopic vacuum geometry.

The standalone mathematical note also needs corrections. Its eigenvalue-7 square-content table says approximately 1.6%; the basis-invariant average over the four-dimensional eigenspace is $1/7 = 14.286\%$. The T2g component alone has zero square content; the A1g component supplies the remainder. Its hex-orbit decomposition in the proof omits T1u, although the conclusion that Eg is absent remains correct. Some uniqueness statements say “every other space-filling cell” despite having checked only the restricted five types. Claims about all monohedral surface-area optima are not established by the supplied spectral argument. #50's printed formula $C_A = F_{hx}/F -$

1 = 3 also fails ordinary substitution (8/14-1 is not 3); an intended alternative normalisation must be written explicitly.

5. Numerical evidence and imported data

Search complexity is real

Paper #72 searches **12,800 total triples**, not 12,800 primitive triples. Its current text explicitly acknowledges that one primitive match in the stated window is consistent with its approximate chance expectation of 1.4. The supplied verification script nevertheless prints a claim of strong statistical evidence from sparsity. That interpretation conflicts with the revised paper. We do not independently certify the approximate null expectation as a calibrated probability model.

The rival-formula script enumerates 1,224,841 distinct rounded values in the family $(a+b \sqrt{17})/(c+d \sqrt{17})$, with integer coefficients between -20 and 20. Re-running it gives these rival counts at its declared tolerances:

Target	Distinct hits	Author's gate
Effective weak angle	102	FAIL
Higgs/Z ratio	620	FAIL
LO solar-angle tangent-squared	12,484	FAIL
Higgs quartic NLO	214	FAIL
Dark-matter/baryon ratio	1,306	FAIL
Baryon asymmetry scaled by 1e10	951	FAIL
Koide ratio	1	PASS

These are reproducible evidence of alternative numerical matches in that chosen grammar. They are **not** calibrated p-values for the whole programme. The gate uses the larger of the quoted uncertainty and achieved residual, and applies arbitrary thresholds (3 and 30 rivals). It does not assign expression-complexity priors or reproduce the complete adaptive search history. Zero rivals for alpha in a grammar that excludes its pi-power formula cannot establish alpha's significance. The same caveat applies when a target or candidate lies outside the grammar.

The README retires the earlier blanket zero-fit evidential framing. This is a substantive improvement. However, it cites a joint-chi-squared defense file and other peer-review deliverables in a directory absent from this public snapshot. Their methodology and conclusions cannot be verified here.

An advertised hydrogen precision uses the measured electron mass

Paper #76 claims the quoted Rydberg result inherits the predicted electron mass. But `explore_unit_map_2026-07.py` directly assigns **me=510998.95 eV**, the empirical value, together with an empirical proton mass. It does not evaluate the displayed electron walk formula. This is a concrete input-lineage problem, not only a concern about symbolic selection.

Evaluating that walk formula with the declared unreduced Planck mass 1.220890e19 GeV gives me approximately **510963.740 eV**, and then Ry approximately **13.60475564 eV**, rather than the advertised 13.605693 eV. Small changes within ordinary Planck-scale uncertainty do not remove a roughly 69-ppm inherited mass discrepancy. The lattice Coulomb tail and approximate bound-state ladder may stand as model calculations, but the displayed high-precision atomic scale is not an independent prediction from the supplied mass formula.

The neutrino reference can omit the solar splitting

If $m_1=0$ in normal ordering, m_3 -squared is Δm_{31} -squared, which equals Δm_{32} -squared plus Δm_{21} -squared. Taking the quoted values $2.453e-3$ and $7.53e-5$ eV-squared gives m_3 approximately **50.2822 meV**. Taking only the square root of the former gives approximately **49.5278 meV**. #72 calls the latter a PDG convention despite using Δm_{32} -squared in its justification. This is not merely a choice of renormalisation convention: the index relation matters. A claimed match at 49.50 meV needs a correctly documented reference and full covariance. #54 explicitly corrects 31 versus 32, but the website still prints the ratio 33 for 32/21.

Other numerical and status corrections

The printed strong coupling on the predictions page is approximately 8.47545; its reciprocal is approximately 0.1179878 and is evidently intended. Recomputed α -inverse is 137.035999054944: the pulls against the stated Cs, Rb and CODATA-2022 values are approximately +0.331, -13.732 and -5.812 respectively. The current homepage fetched here does not show the earlier alleged 1.4-sigma Rb statement, so that criticism should not be attributed to the current homepage. A raw pull against a selected determination is not a theory-level discovery significance.

Paper #17 explicitly starts its running calculation with observed couplings, including α -s, then advertises α -s reproduced exactly. That is not an independent forward prediction. Its GUT coupling $1/46$ also differs from the $1/25$ used elsewhere. Paper #59 assumes three active quark flavors at the Z scale and omits the Higgs contribution in its quoted weak beta coefficient; flavors, schemes and thresholds must be specified consistently. Standard QCD running is documented in the PDG 2025 QCD review: <https://pdg.lbl.gov/2025/reviews/rpp2025-rev-qcd.pdf>.

Paper #55's printed tilt formula evaluates at $r=0.0225$ to **n_t approximately -0.001004**, not -0.008. Its replacement of r approximately 0.063 by approximately 0.0225 followed known observational tension; the revised form can be frozen for future tests but the revision is model selection, not fresh confirmation by the pre-existing limit. The 0.0225 number uses observed n_s ; its own n_s gives a different value near 0.0243.

6. Other claims needing substantial correction

Strong CP and hierarchy (#32–33): choosing a torsion potential with its minimum at zero is not a derivation of the physical QCD theta angle. Mass rephasings and the anomalous theta shift must be handled. Theta=0 alone does not imply exactly zero total neutron EDM when CKM CP violation remains. A finite physical UV cutoff removes an infinite integral, but not large finite contributions proportional to the cutoff squared; it does not by itself solve Higgs naturalness.

Confinement/topological conservation (#14, #18): an irrep or vector space does not define a vacuum manifold whose fundamental group is $\mathbb{Z}$. The manifold, energy functional and allowed finite-energy configurations need to be specified. Topological winding restrictions do not by themselves establish QCD confinement or a string tension. Exact separate baryon/lepton conservation must be reconciled with the intended electroweak anomaly physics and baryogenesis in #49/#61, or an explicit primordial asymmetry premise.

Cosmology (#49, #53, #55, #61, #70): counting projector dimensions does not calculate vacuum stress-energy. A stationary band edge has zero group velocity but need not have zero energy; this undermines the “non-wave-energy modes” reading used for 6/7. An observed radius in a Planck-to-cosmological density scaling is an empirical input. Dark-matter abundance requires evolution and perturbation dynamics; reproducing a background ratio does not establish structure formation, lensing or CMB power spectra. Baryogenesis requires a thermal history, CP

source, rates and transport, not only a dimensionless combination close to η . The first-order electroweak transition needs a potential and finite-temperature calculation compatible with the proposed particle content.

Quantum reconstruction (#2, #45, #75): the Cauchy/additivity argument for Born weights is meaningful conditional mathematics. The supplied verifiers acknowledge that microscopic measurement premises and the laboratory-qubit identification remain open. Stationarity can remove a phase/time-origin dependence; it does not fix a quadratic amplitude dependence to all orders for arbitrary detector thresholds. A normalized entangled state and measurement probabilities already in Hilbert space reproduce Bell correlations; this is not an independent emergence of that physical framework. No-signalling must be established for the allowed preparations and interventions, not inferred from bulk incompressibility alone. The withdrawal of the earlier finite-speed void mechanism is a responsible correction and should remain clearly visible.

Periodic table and visible-spectrum claims (#30, #37): familiar angular-momentum degeneracies and crystal-field decompositions are not a derivation of all many-electron ordering, screening and exceptions. #37 equates the dimensionless value 40.5 with hertz, chooses octave counts, and links this to named colors, tuning systems and pineal crystals; the frequency scale and transduction law are not supplied. The paper acknowledges several gaps and labels itself speculative. It should be excluded from the evidence for fundamental unification. The presence of seven named colors is not an independently measured seven-mode spectral discretisation.

7. Prospective tests: preserve them, but make them operational

Hypothesis	What must be frozen before testing
PMNS/CKM phase ratio equals 3	Mixing-matrix convention, circular-angle branches, LO versus revised CKM phase, joint covariance and a rule for future corrections. Current advertised individual phase values do not all maintain the exact ratio.
$m_1=0$, normal ordering, mass sum near 58 meV	Correct splitting definitions, reference fit, uncertainty propagation and cosmological model assumptions. Normal ordering alone is weak discrimination among many models.
r relation near 0.022–0.024	Pivot wavenumber, relation to ns, corrected tensor tilt, theoretical uncertainty and prohibition of post-result formula changes.
Altitude decoherence	A specified experimental observable, coordinate versus proper-time rates, noise/background controls and detectable uncertainty. A tiny null effect without adequate sensitivity is not falsification.
No DM particle, no SUSY, no axion	State the ontology that excludes them. Positive identification could refute it; finite null searches only exclude the tested parameter regions.
Zero neutron EDM, absolute proton stability, no neutrinoless double-beta decay	Resolve contributions from the intended SM sector and state exact scope. No finite experiment confirms exact zero or infinite lifetime.

Hypothesis	What must be frozen before testing
Quadratic Lorentz violation	Coefficients, directions, sectors, common-speed assumptions and a consistently derived Hermitian dispersion. Merely giving a suppression order is not a sharp numerical prediction.

“Pre-registered” needs an immutable timestamped statement of the exact formula and acceptance/rejection rule. A current repository label or a project’s general priority date is insufficient to demonstrate what was frozen before which data. We have not verified preregistration provenance for all advertised tests.

8. Required revisions before another physical referee round

1. Withdraw the claim that #59–60 currently close SM+GR; correct the explicit errors above and propagate those status changes through all dependent papers, the book, scripts and website.
2. Publish one complete microscopic model, including time, inter-cell coupling, measure, interactions, units and every physical premise. Supply a dependency table distinguishing proven identities from assumed identifications.
3. Rebuild continuum matching with a Hermitian operator and genuine spinor/gauge/gravity structures. Check gauge anomalies, chirality, locality and limiting speeds with the complete matter content.
4. Separate observed inputs, retrospective formula choices and prospective outputs. Restore the missing public joint-statistics deliverables and show complete input lineage, including the hydrogen mass substitution.
5. Freeze a small operational set of future predictions, versioned independently of later revisions. Use correlated likelihoods and transparent theoretical errors, not sums of selectively quoted sigma residuals.
6. Submit the strongest standalone graph results with corrected tables and precise scope. Investigate novelty rather than infer it from physical enthusiasm.

An appropriate future statistical audit would enumerate the actual discovery grammar, quotient equivalent expressions, account for data-dependent corrections and calculate description lengths. It would evaluate a predeclared joint set under null targets with realistic scales and covariances. The present rival gate is useful diagnostic evidence but is not that full audit. No calibrated global false-discovery probability is claimed here.

9. Corrections to the earlier conversational review

The earlier review’s overall caution is supported, but its numerical ratings were subjective and should not be treated as refereeing evidence. This report replaces them with explicit defects and reproducible checks. In particular:

- The neutrino enumeration contains 12,800 total triples, not 12,800 primitive triples.
- The current homepage does not contain the previously attributed Rb 1.4-sigma line.
- The restricted Fedorov characterisation does not certify surface-area optimality, vacuum geometry or novelty.
- The sharpest new objections are algebraic, code-level and input-lineage failures, not merely “missing physical bridges.”

- Reproducible structural mathematics and responsible retractions should receive credit, without transferring that credit to unproved dependent physical claims.

Source locator and reproduction

Primary UFFT source: <https://github.com/ufft-info/UFFT> at the commit above. Current web pages checked: <https://ufft.info/> and <https://ufft.info/predictions>. Every inventory row includes its archive and repository URLs. Direct source snapshots, hashes and execution logs are retained in the companion package.

Important passages: #59 §§2, 3.3, 4.2, 5.1, 6.1, 6.4; #60 §§2.1, 4.2–4.4; #57 v2 changes and §3; #58 v2 §§5–8; #72 §§3–6; #55 §§5–9; #76 §5; standalone Fedorov note §§6 and 8.2. Code: `Symanzik_Matching_BCC.py` Bloch builder and eigvalsh calls; `explore_unit_map_2026-07.py` headline function; `MatchGate_2026-07.py`; `verify_Paper72_0h_irreps.py` V11 interpretation.

Independent check: run `python independent_checks.py` with Python 3 and NumPy. In the supplied package it also checks the frozen Symanzik builder. Saved results are in `evidence/independent_results.json`. Author runs use their frozen code and their quoted historical experimental references; this audit does not certify that those are the latest global fits.

Final referee judgment: the central physical conclusion is unsupported by the present public proof. Some subsidiary arguments are demonstrably incorrect as written. Retain the graph results and conditional constructions; rebuild the physical derivation and evidence claims before seeking endorsement of unification.

Paper-by-paper screening register

This appendix preserves all 81 indexed paper/version rows from the audit register. **D** indicates selected detailed argument or code checks, not complete proof certification. **S** indicates principal-claim screening; full derivation and current empirical-fit audits remain incomplete. Source titles and indexed claims are the author's labels, not accepted findings. Repeated versions and cross-listed entries remain distinct.

Entry 1: Face Laplacian Spectrum

Paper/version: 5. **Audit depth:** D.

Indexed claim: $\text{Spec}(L) = \{0^1, (9 - \sqrt{17})/2^3, 4^2, (9 + \sqrt{17})/2^3, 7^4, 9^1\}$

Referee verdict: Confirmed graph result.

Assumptions, selection and testing gaps: Exact polynomial independently reconstructed; physical meaning and novelty are separate.

Archive locator: <https://zenodo.org/records/19030062>.

Entry 2: Master Equation

Paper/version: 16. **Audit depth:** S.

Indexed claim: $\lambda^2 - 9\lambda + 16 = 0, \Delta = 17, C_A = 3$

Referee verdict: Graph identity plus interpretation.

Assumptions, selection and testing gaps: Quadratic factor valid; C_A identification and universal physical readout do not follow from it.

Archive locator: <https://zenodo.org/records/19064359>.

Entry 3: Local Decision Dynamics

Paper/version: ?. **Audit depth:** S.

Indexed claim: Local decision principle, 8 structural results

Referee verdict: Unnumbered hypothesis/correction.

Assumptions, selection and testing gaps: Local decision mapping unproved; speed withdrawal should be preserved and dependent claims versioned.

Archive locator: <https://zenodo.org/records/19363008>.

Entry 4: SM From One Matrix

Paper/version: 48. **Audit depth:** D.

Indexed claim: $S = \psi^\dagger L_T \psi \rightarrow \text{SM Lagrangian}; v_2 = \text{Paper \#72 integration} + \text{abstract rewrite}$

Referee verdict: Conjecture with admitted fits.

Assumptions, selection and testing gaps: Latest abstract acknowledges α_s fit and baryogenesis post-hoc factor; continuum action incomplete.

Archive locator: <https://zenodo.org/records/19662029>.

Archive locator: <https://zenodo.org/records/19448024>.

Entry 5: Uniqueness of Foam Cell**Paper/version:** 50. **Audit depth:** D.

Indexed claim: Trunc oct unique among Fedorov cells ($\Delta = 17$ prime); v2 = criterion (iii) sharpened + Weaire-Phelan scope note. Independent confirmation (Oct 2026): the face graph is the tetrakis hexahedral graph, whose tabulated characteristic polynomial has discriminant 17; Core Part X §10

Referee verdict: Restricted graph characterisation.

Assumptions, selection and testing gaps: Five-type spectral selection reproduced for supplied graphs; not global minimum-area theorem. Printed C_A definition incorrect.

Archive locator: <https://zenodo.org/records/19662068>.**Archive locator:** <https://zenodo.org/records/19447996>.**Entry 6: Central Theorem****Paper/version:** 59. **Audit depth:** D.

Indexed claim: Foam $\rightarrow$ SM+GR in five steps; v2 (2026-07-12) = torsion citations corrected (T_hex vs T), composite status = proof-sketch pending external audit

Referee verdict: Blocking defects.

Assumptions, selection and testing gaps: Gauge links imported, invalid displayed gamma construction, zero Yukawa projector product, Higgs sign error, unreliable matching.

Archive locator: <https://zenodo.org/records/21323529>.**Archive locator:** <https://zenodo.org/records/19491095>.**Entry 7: Four Closing Theorems****Paper/version:** 60. **Audit depth:** D.**Indexed claim:** Anomaly, generations, gravity, continuum completeness**Referee verdict:** Blocking defects.

Assumptions, selection and testing gaps: Nonzero anticommutator claimed zero; GR elasticity/Weinberg-Witten misuse; no completeness closure.

Archive locator: <https://zenodo.org/records/19491125>.**Entry 8: Pure Mathematics of the Kelvin Cell****Paper/version:** 63. **Audit depth:** D.

Indexed claim: Face Laplacian spectrum, O_h irreps, $Q(\sqrt{17})$ ring, Fedorov uniqueness (Tier 1 standalone)

Referee verdict: Graph mathematics with corrections.

Assumptions, selection and testing gaps: Exact spectrum valid; generation/energy meanings external; restricted uniqueness scope and face-content statements need checking.

Archive locator: <https://zenodo.org/records/19624955>.

Entry 9: Fine Structure Constant v1

Paper/version: 3. **Audit depth:** D.

Indexed claim: $\alpha^{-1} = 137.035999055$ (0.21 ppb)

Referee verdict: Postdiction.

Assumptions, selection and testing gaps: Finite series coefficients, normalization and truncation need unique physical derivation; conflicting recoil measurements must all be reported.

Archive locator: <https://zenodo.org/records/19011758>.

Entry 10: Fine Structure Constant v2

Paper/version: 3 (v2). **Audit depth:** D.

Indexed claim: Uniqueness proof, CW-complex expansion

Referee verdict: Not inspected: indexed file absent.

Assumptions, selection and testing gaps: Finite series coefficients, normalization and truncation need unique physical derivation; conflicting recoil measurements must all be reported.

Archive locator: <https://zenodo.org/records/19019944>.

Access limit: the indexed Markdown file was absent from the frozen repository snapshot.

Entry 11: Fine Structure Constant v3

Paper/version: 3 (v3). **Audit depth:** D.

Indexed claim: Updated with running coupling

Referee verdict: Not inspected: indexed file absent.

Assumptions, selection and testing gaps: Finite series coefficients, normalization and truncation need unique physical derivation; conflicting recoil measurements must all be reported.

Archive locator: <https://zenodo.org/records/19063910>.

Access limit: the indexed Markdown file was absent from the frozen repository snapshot.

Entry 12: Fine Structure Constant v4

Paper/version: 3 (v4). **Audit depth:** D.

Indexed claim: Complete derivation, 101 views, 23 downloads

Referee verdict: Postdiction.

Assumptions, selection and testing gaps: Finite series coefficients, normalization and truncation need unique physical derivation; conflicting recoil measurements must all be reported.

Archive locator: <https://zenodo.org/records/19308917>.

Entry 13: D-Mode Path Integral

Paper/version: 22. **Audit depth:** S.

Indexed claim: α as heat kernel Z_D on face graph

Referee verdict: Unestablished bridge.

Assumptions, selection and testing gaps: Graph heat-kernel calculation does not establish electromagnetic normalization or the selected expansion.

Archive locator: <https://zenodo.org/records/19084565>.

Entry 14: g-2 Schwinger Term

Paper/version: 21. **Audit depth:** S.

Indexed claim: $(g-2)/2 = \alpha / (2 \pi)$ from foam

Referee verdict: Conditional.

Assumptions, selection and testing gaps: Schwinger coefficient relies on QED/continuum identifications; loop dynamics not independently certified.

Archive locator: <https://zenodo.org/records/19080011>.

Entry 15: g-2 Two-Loop C_2

Paper/version: 23. **Audit depth:** S.

Indexed claim: C_2 structural identification

Referee verdict: Superseded/conditional.

Assumptions, selection and testing gaps: Two-loop structural identifications must be read with later corrigendum and #68 obstruction.

Archive locator: <https://zenodo.org/records/19084710>.

Entry 16: g-2 C_2 Complete

Paper/version: 24. **Audit depth:** S.

Indexed claim: Full 2-loop derivation

Referee verdict: Conditional/imported.

Assumptions, selection and testing gaps: QED loop integrals and transcendental terms need forward lattice derivation; algebraic matching insufficient.

Archive locator: <https://zenodo.org/records/19084873>.

Entry 17: C_2 Corrigendum

Paper/version: 26. **Audit depth:** S.

Indexed claim: Independent derivation, 3×10^{-16} agreement

Referee verdict: Correction/conditional.

Assumptions, selection and testing gaps: A corrected numerical coefficient is useful but remains dependent on the continuum-QED bridge.

Archive locator: <https://zenodo.org/records/19085997>.

Entry 18: Rational Term Closure**Paper/version:** 27. **Audit depth:** D.**Indexed claim:** 197/144 derived, α uniqueness proven**Referee verdict:** Identity/imported.**Assumptions, selection and testing gaps:** 197/144 obtained through QED intermediary, admitted in #68; coefficient uniqueness not established by symmetry alone.**Archive locator:** <https://zenodo.org/records/19102302>.**Entry 19: Cell-Integer Identities and Single-Cell Obstruction to 197/144****Paper/version:** 68. **Audit depth:** D.**Indexed claim:** T68.1 $(F-1)(F-14)=0$; T68.2 $\text{hex-hex}=12$, $\text{tr}(L_1^2)=288$, $\text{tr}(A_{\text{face}}^3)=144$; T68.3 $Q(\sqrt{17})$ recurrence rational/transcendental split; T68.4 single-cell obstruction ($\min L^1 = 1/144$)**Referee verdict:** Useful identities/negative result.**Assumptions, selection and testing gaps:** Admits imported QED rational; obstruction concerns a specified single-cell grammar, not all multi-cell actions.**Archive locator:** <https://zenodo.org/records/19658979>.**Entry 20: Weinberg Angle v1 (GUT)****Paper/version:** 11. **Audit depth:** S.**Indexed claim:** $\sin^2 \theta_W = 3/8$ at GUT scale**Referee verdict:** Conditional/imported.**Assumptions, selection and testing gaps:** $SU(5)$ fundamental and hypercharge normalization assumed; RG inputs are not a derived gauge embedding.**Archive locator:** <https://zenodo.org/records/19063822>.**Entry 21: Weinberg Angle v2****Paper/version:** 41. **Audit depth:** S.**Indexed claim:** $\sin^2 \theta_W = (17-3\sqrt{17})/20 = 0.23153$ **Referee verdict:** Postdiction/open derivation.**Assumptions, selection and testing gaps:** Renormalization scheme/scale required; #58 v2 explicitly leaves mixing weight open.**Archive locator:** <https://zenodo.org/records/19306610>.**Entry 22: Higgs/Z Mass Ratio****Paper/version:** 13. **Audit depth:** S.**Indexed claim:** $m_H/M_Z = 18/(9+\sqrt{17})$, 0.14%**Referee verdict:** Postdiction.

Assumptions, selection and testing gaps: Eigenvalue-to-mass mapping selected; 620 rival numerical matches in the tested grammar.

Archive locator: <https://zenodo.org/records/19064036>.

Entry 23: Electroweak Predictions

Paper/version: 19. **Audit depth:** S.

Indexed claim: M_W/M_Z , $N_\nu=3$, $d=3+1$, $\rho=1$

Referee verdict: Conditional.

Assumptions, selection and testing gaps: EW content, mass relations and Lorentz structures depend on the unsupported assignments.

Archive locator: <https://zenodo.org/records/19064635>.

Entry 24: Necessity of Standard Model

Paper/version: 57. **Audit depth:** D.

Indexed claim: $\text{Higgs}=A_2u$, chiral structure forced; v_2 (2026-07-12) = T_{hex} /inter-type T split, 57.1 premises corrected, 57.2 localisation claim withdrawn (50/50 bands, labelling conditional)

Referee verdict: Partial correction/unproved necessity.

Assumptions, selection and testing gaps: Corrected annihilation and equal-band weights; negative charge-to-mass rule is a premise, not SSB theorem.

Archive locator: <https://zenodo.org/records/21323321>.

Archive locator: <https://zenodo.org/records/19484509>.

Entry 25: Gauge Sector Placement

Paper/version: 58. **Audit depth:** D.

Indexed claim: $SU(3) \times SU(2) \times U(1)$ forced by exhaustion; v_2 (2026-07-12) = Thm 58.3 demoted to Result 58.3 (Tier 2, derivation open), exact identity stated, T_{2g} content exactly 100% hex

Referee verdict: Conditional placement/open bridge.

Assumptions, selection and testing gaps: SU_2 embedding statement wrong for S_3 ; weak angle admitted open; placement criteria presuppose physical content.

Archive locator: <https://zenodo.org/records/21323498>.

Archive locator: <https://zenodo.org/records/19484967>.

Entry 26: Asymptotic Freedom & Confinement

Paper/version: 18. **Audit depth:** S.

Indexed claim: $b_0=7=\lambda_{T_{2g}}$, confinement from $\pi_1=\mathbb{Z}$

Referee verdict: Unproved physical theorem.

Assumptions, selection and testing gaps: Ordinary beta function imported; alleged winding confinement needs manifold, action and energy bound.

Archive locator: <https://zenodo.org/records/19064581>.

Entry 27: GUT Scale Beta Functions

Paper/version: 17. **Audit depth:** D.

Indexed claim: $M_{\text{GUT}}=1.50 \times 10^{16} \text{ GeV}$, $\alpha_s=0.1179$

Referee verdict: Input-dependent.

Assumptions, selection and testing gaps: Observed α_s supplied as boundary input then reproduced; GUT coupling differs across sources.

Archive locator: <https://zenodo.org/records/19064416>.

Entry 28: CKM Matrix

Paper/version: 34. **Audit depth:** S.

Indexed claim: $\lambda = \sin(\pi/14)$, $A=r_1/C_A$

Referee verdict: Postdiction.

Assumptions, selection and testing gaps: Angle and texture mapping selected; LO residual and quoted sigma arithmetic need consistent inputs.

Archive locator: <https://zenodo.org/records/19198360>.

Entry 29: PMNS Matrix

Paper/version: 35. **Audit depth:** S.

Indexed claim: $\tan^2 \theta_{12} = \sqrt{17}/9$, $\sin^2 \theta_{13} = (\sqrt{17}/27)^2$

Referee verdict: Postdiction/superseded.

Assumptions, selection and testing gaps: LO mixing relations later corrected; 31 versus 32 mass-splitting ratio must be distinguished.

Archive locator: <https://zenodo.org/records/19198422>.

Entry 30: CP Phases

Paper/version: 36. **Audit depth:** S.

Indexed claim: $\delta_{\text{PMNS}}/\delta_{\text{CKM}} = 3 \text{ exactly}$ (DUNE test)

Referee verdict: Hypothesis with future test.

Assumptions, selection and testing gaps: Phase ratio test useful only with frozen convention/branches and consistent updated individual phases.

Archive locator: <https://zenodo.org/records/19198775>.

Entry 31: NLO Mixing Correction

Paper/version: 51. **Audit depth:** S.

Indexed claim: $\epsilon = \sqrt{17}/81$, Cabibbo $3.7 \sigma \rightarrow 0.07 \sigma$

Referee verdict: Unproved uniqueness/postselection.

Assumptions, selection and testing gaps: Z2-odd symmetry does not uniquely fix denominator/power of a perturbation; explicit physical matching needed.

Archive locator: <https://zenodo.org/records/19477100>.

Entry 32: Wolfenstein $\bar{\rho}$ NLO

Paper/version: 64. **Audit depth:** S.

Indexed claim: $\bar{\rho} = 0.159$ (0.002σ), $\text{tr}(H) = 1/3$, $\det(H) = -8$

Referee verdict: Postdiction.

Assumptions, selection and testing gaps: Wolfenstein correction and normalization require pre-data unique derivation; joint CKM covariance missing.

Archive locator: <https://zenodo.org/records/19624977>.

Entry 33: Wolfenstein A NLO

Paper/version: 66. **Audit depth:** S.

Indexed claim: $A = (19 + \sqrt{17})/28 = 0.826$ (0.015σ)

Referee verdict: Postdiction.

Assumptions, selection and testing gaps: A expression near fit is not uniquely selected by irrep symmetry; use joint Wolfenstein covariance.

Archive locator: <https://zenodo.org/records/19625071>.

Entry 34: NLO $\delta_{\text{CKM}} / \bar{\eta}$ Closure

Paper/version: 67. **Audit depth:** S.

Indexed claim: $\delta_{\text{NLO}} = \delta_{\text{LO}} \times 71/72 = 65.438^\circ$ ($\pm 0.002^\circ$ from PDG); $R_b = 13/34$

Referee verdict: Postdiction/version mismatch.

Assumptions, selection and testing gaps: 71/72 correction and R_b choice selected; updated CKM phase must preserve or revise advertised exact PMNS ratio.

Archive locator: <https://zenodo.org/records/19625122>.

Entry 35: R_b NLO Denominator (Fermion-Walk Operator PT)

Paper/version: 69. **Audit depth:** S.

Indexed claim: $2V-F = \dim(H_{\text{fermion}}^{\text{eff}})$; joint ($\bar{\rho}, \bar{\eta}$) at 0.02σ vs PDG; closes Paper #67 R_b companion at Tier 2

Referee verdict: Conditional/postdiction.

Assumptions, selection and testing gaps: Perturbation denominator and effective-state count require a physical operator derivation; website and index give differing R_b forms.

Archive locator: <https://zenodo.org/records/19626475>.

Entry 36: PMNS Solar Angle NLO (Eigenvalue Self-Energy)**Paper/version:** 71. **Audit depth:** S.**Indexed claim:** $\tan^2 \theta_{12} = (\sqrt{17}/9)(1 - \sqrt{17}/144) = 0.44501$ (+0.074 σ vs PDG), $7.5 \times$ tightening over LO; closes last $>0.3 \sigma$ PMNS residual; Tier 2**Referee verdict:** Postdiction/unproved self-energy.**Assumptions, selection and testing gaps:** Symmetry alone does not fix shift magnitude; explicit loop action, normalization and scheme needed.**Archive locator:** <https://zenodo.org/records/19626978>.**Entry 37: Lepton Mass Ratios v1****Paper/version:** 10. **Audit depth:** S.**Indexed claim:** Koide $Q=2/3$ as theorem**Referee verdict:** Unestablished physical theorem.**Assumptions, selection and testing gaps:** Koide parametrization algebra conditional; symmetry alone does not fix its amplitude or physical masses.**Archive locator:** <https://zenodo.org/records/19063774>.**Entry 38: Lepton Mass Ratios v2****Paper/version:** 31. **Audit depth:** S.**Indexed claim:** Koide $\theta = 2/9$, 10-70 ppm**Referee verdict:** Postdiction.**Assumptions, selection and testing gaps:** Koide angle selected from spectral ratios; physical mass relation is not forced by irrep symmetry.**Archive locator:** <https://zenodo.org/records/19185685>.**Entry 39: Complete Particle Masses****Paper/version:** 44. **Audit depth:** S.**Indexed claim:** All 13 masses, zero fitted parameters**Referee verdict:** Postdiction.**Assumptions, selection and testing gaps:** Flavor-specific integer expressions selected; mass schemes and observed dimensional-scale inputs need explicit provenance.**Archive locator:** <https://zenodo.org/records/19307003>.**Entry 40: NLO Neutrinos, α_s , M_W** **Paper/version:** 47. **Audit depth:** S.**Indexed claim:** $m_1=0$, $m_3=49.5$ meV, $\alpha_s=0.11799$ **Referee verdict:** Postdiction/conditional.**Assumptions, selection and testing gaps:** NLO choices, α_s flavor condition and neutrino reference need audit; zero lightest mass assumes selected mass matrix.

Archive locator: <https://zenodo.org/records/19448066>.

Entry 41: n-p Mass Difference NLO

Paper/version: 52. **Audit depth:** S.

Indexed claim: $\Delta m = 1.2933 \text{ MeV} (0.34 \sigma)$

Referee verdict: Postdiction.

Assumptions, selection and testing gaps: NLO denominator selected; correct QCD+EM mass schemes, covariance and physical correction needed.

Archive locator: <https://zenodo.org/records/19477132>.

Entry 42: Neutrino Mass Ratio

Paper/version: 54. **Audit depth:** S.

Indexed claim: $\Delta m^2_{31} / \Delta m^2_{21} = 33$ exactly

Referee verdict: Identity plus hypothesis.

Assumptions, selection and testing gaps: Frobenius identity 33 valid; mapping to Delta31/Delta21 not derived; website still gives Delta32 ratio.

Archive locator: <https://zenodo.org/records/19484047>.

Entry 43: Neutrino Mass Ratio — Eisenstein Norm

Paper/version: 65. **Audit depth:** S.

Indexed claim: $33 = N(r_1 + r_2 \omega) = S^2 - C_A \cdot P$, Tier 4 $\rightarrow$ 2 promotion

Referee verdict: Identity plus hypothesis.

Assumptions, selection and testing gaps: Eisenstein norm can equal 33; identification with neutrino splittings and inherited reference need a physical derivation.

Archive locator: <https://zenodo.org/records/19625005>.

Entry 44: Dirac Operator, Generation Count, Chirality, m_3 Integer

Paper/version: 72. **Audit depth:** D.

Indexed claim: T72.1 $D_F^2 = L_T$, T72.2 $N_{\text{gen}} = \text{mult}(T_1 u) = 3$, T72.3a L_{off}

Referee verdict: Mixed: valid blocks, invalid inference.

Assumptions, selection and testing gaps: $T_1 u$ multiplicity two not three; square-root not relativistic Dirac proof; 12,800-total search matches chance expectation.

Archive locator: <https://zenodo.org/records/19658759>.

Entry 45: Lorentz γ from Foam Wave Kinematics

Paper/version: 73. **Audit depth:** D.

Indexed claim: First kinematic γ derivation ($\square \phi$ substitution); four corollaries; linearized moving-Schwarzschild + Kerr dipole; $\delta c/c$ O_h-quartic cross-check

Referee verdict: Conditional relativistic mathematics.

Assumptions, selection and testing gaps: Boost relation follows assumed continuum wave operator; microscopic Lorentz invariance not established.

Archive locator: <https://zenodo.org/records/21323677>.

Entry 46: Structural Theorems for the Electron Walk Action

Paper/version: 74. **Audit depth:** D.

Indexed claim: Three Tier-1 structural theorems on S_{walk} (topological step count, T_{1u} spectral factorisation, colour-group denominator); C74.4 path-integral conjecture stated

Referee verdict: Identities plus action conjecture.

Assumptions, selection and testing gaps: Topological/spectral rewritings do not establish physical walk weighting or selected mass exponent.

Archive locator: <https://zenodo.org/records/21323789>.

Entry 47: The Born Rule from Imprint Statistics and the Entangled Pair State from the Antiunitary Twin Map

Paper/version: 75. **Audit depth:** D.

Indexed claim: Born rule derived from phase-blindness (substrate stationarity) + fine-graining additivity (no-signalling, P#45 §8); twin state Σ

Referee verdict: Conditional quantum theorem.

Assumptions, selection and testing gaps: Born additivity and antiunitary-twin construction useful under premises; detectors and full microscopic premise discharge unproved.

Archive locator: <https://zenodo.org/records/21323993>.

Entry 48: Hydrogen from the Foam

Paper/version: 76. **Audit depth:** D.

Indexed claim: $1/r$ from adjacency ($A=1/(4\pi)$ exact), charge = A1g dressing, $SO(4)$ ladder, $R_y = \alpha^2 m_e c^2 / 2 = 13.605693$ eV (no new parameters), $\delta = -K/a_B^2$ with $K = 3.810$ closed form

Referee verdict: Conditional lattice results/input error.

Assumptions, selection and testing gaps: Hydrogen unit-map code uses observed electron mass instead of walk prediction; advertised precision not inherited from proposed model.

Archive locator: <https://zenodo.org/records/21339706>.

Entry 49: Vacuum Metric

Paper/version: 7. **Audit depth:** S.

Indexed claim: Schwarzschild + Kerr from foam

Referee verdict: Circular/conditional.

Assumptions, selection and testing gaps: Covariance, vacuum Einstein equations or metric/null assumptions used to reconstruct GR solutions.

Archive locator: <https://zenodo.org/records/19063610>.

Entry 50: Friedmann & Einstein-Hilbert

Paper/version: 9. **Audit depth:** D.

Indexed claim: Both Friedmann eqs from foam

Referee verdict: Conditional.

Assumptions, selection and testing gaps: Covariance, Newtonian gravity and relativistic continuity imported; Planck-unit definitions do not predict G.

Archive locator: <https://zenodo.org/records/19063718>.

Entry 51: Schwarzschild Complete

Paper/version: 28. **Audit depth:** S.

Indexed claim: g_{rr} from incompressibility $\nu = 1/2$

Referee verdict: Conditional.

Assumptions, selection and testing gaps: Incompressibility-to-metric mapping and covariance supplied; reconstructing Schwarzschild does not prove GR.

Archive locator: <https://zenodo.org/records/19184702>.

Entry 52: Kerr Metric

Paper/version: 46. **Audit depth:** S.

Indexed claim: All 5 components, spin-1/2 from Wilson loop

Referee verdict: Conditional reconstruction.

Assumptions, selection and testing gaps: Kerr-like density, determinant and torsion mappings supplied; spin statistics do not follow merely from a Wilson-loop phase.

Archive locator: <https://zenodo.org/records/19307177>.

Entry 53: Lorentz γ from Foam Wave Kinematics

Paper/version: 73. **Audit depth:** D.

Indexed claim: $\gamma^2(1-v^2/c^2)=1$ from $\square$ substitution (first kinematic derivation); $L=L_0/\gamma$, $\tau = \gamma \tau_0$, SO(3,1) closure; linearized moving-Schwarzschild + Kerr dipole; $\delta c/c$ O_h-quartic cross-check

Referee verdict: Conditional relativistic mathematics.

Assumptions, selection and testing gaps: Boost relation follows assumed continuum wave operator; microscopic Lorentz invariance not established.

Archive locator: <https://zenodo.org/records/21323677>.

Entry 54: Maxwell Equations v1

Paper/version: 8. **Audit depth:** S.

Indexed claim: All 4 Maxwell eqs from $\square D=0$

Referee verdict: Conditional.

Assumptions, selection and testing gaps: Potential, sources and covariance supplied; a wave equation alone does not fix all sourced Maxwell dynamics.

Archive locator: <https://zenodo.org/records/19063671>.

Entry 55: Maxwell Equations Full

Paper/version: 29. **Audit depth:** S.

Indexed claim: Charge quantisation from O_h

Referee verdict: Conditional.

Assumptions, selection and testing gaps: Helmholtz/Volterra and charge-defect identification do not fix sourced EM action without further premises.

Archive locator: <https://zenodo.org/records/19185556>.

Entry 56: Decoherence Suppression

Paper/version: 1. **Audit depth:** S.

Indexed claim: $\Delta \Gamma / \Gamma = 8.22 \times 10^{-11}$

Referee verdict: Hypothesis.

Assumptions, selection and testing gaps: Vacuum-density-to-decoherence mapping and clock convention; altitude experiment needs a sensitivity budget.

Archive locator: <https://zenodo.org/records/18706756>.

Entry 57: Bell Correlations (v2.0)

Paper/version: 2. **Audit depth:** D.

Indexed claim: Void-pair conservation $B + V = D$ as the origin of Bell correlations; singlet state and CHSH; v2.0 supersedes 18706806 (§4.1 corrected per Paper #75)

Referee verdict: Conditional.

Assumptions, selection and testing gaps: Singlet/Hilbert-space and imprint premises supply correlations; earlier mechanism superseded. Microscopic no-signalling remains to be shown.

Archive locator: <https://zenodo.org/records/21331868>.

Archive locator: <https://zenodo.org/records/18706806>.

Entry 58: Void Network Geometry and Bell Non-Locality: Withdrawal of the $c \sqrt{3/2}$ Propagation Speed

Paper/version: ?. **Audit depth:** S.

Indexed claim: v1.0 result ($c_V = c \sqrt{3/2}$) WITHDRAWN: void spacing is $a/2$, front-speed ratio ≤ 1 , finite speeds excluded (Salart 2008, Bancal 2012). Mechanism: Paper #45 void channel

Referee verdict: Unnumbered hypothesis/correction.

Assumptions, selection and testing gaps: Local decision mapping unproved; speed withdrawal should be preserved and dependent claims versioned.

Archive locator: <https://zenodo.org/records/23157808>.

Archive locator: <https://zenodo.org/records/19079502>.

Entry 59: S-Matrix & LSZ

Paper/version: 25. **Audit depth:** S.

Indexed claim: LSZ from foam, Compton 0.002%

Referee verdict: Conditional.

Assumptions, selection and testing gaps: LSZ assumes continuum QFT, asymptotic states and field normalization; cannot certify their emergence.

Archive locator: <https://zenodo.org/records/19085007>.

Entry 60: Baryon/Lepton Conservation

Paper/version: 14. **Audit depth:** S.

Indexed claim: $\pi_1(T_2g) = \mathbb{Z}$, proton stable

Referee verdict: Unproved topology.

Assumptions, selection and testing gaps: Vacuum manifold/finite-energy action absent; separate B,L conservation requires reconciliation with EW anomalies and baryogenesis.

Archive locator: <https://zenodo.org/records/19064093>.

Entry 61: Inter-Type Torsion Operator

Paper/version: 39. **Audit depth:** S.

Indexed claim: $\delta_{CKM} = 66.36^\circ$, $\sin^2 \theta_W = 0.2315$

Referee verdict: Conditional/postdiction.

Assumptions, selection and testing gaps: Geometry/torsion complex phase not yet a physical CP invariant; compare later definitions and corrections.

Archive locator: <https://zenodo.org/records/19306447>.

Entry 62: Total Torsion Identity

Paper/version: 40. **Audit depth:** S.

Indexed claim: $2\theta_{sh} + \theta_{hh} = \pi$, spin-1/2

Referee verdict: Conditional mathematics.

Assumptions, selection and testing gaps: Dihedral identity plausible; spin/statistics and Planck-time normalization are extra physical identifications.

Archive locator: <https://zenodo.org/records/19306543>.

Entry 63: Torsion T_1u Theorems

Paper/version: 56. **Audit depth:** S.

Indexed claim: $T^2 = -4I$, maximal generation symmetry

Referee verdict: Conditional graph mathematics.

Assumptions, selection and testing gaps: T1u torsion identities support selected graph blocks; scalar generation coupling does not generate flavor hierarchy.

Archive locator: <https://zenodo.org/records/19484354>.

Entry 64: Strong CP Dissolved

Paper/version: 32. **Audit depth:** S.

Indexed claim: $\theta = 0$ from torsion, $d_n = 0$, no axion

Referee verdict: Unproved/excessive.

Assumptions, selection and testing gaps: Torsion potential chosen; anomalous mass rephasing unresolved; $\theta = 0$ does not force zero total neutron EDM.

Archive locator: <https://zenodo.org/records/19196872>.

Entry 65: Hierarchy Problem Dissolved

Paper/version: 33. **Audit depth:** S.

Indexed claim: No quadratic divergence on lattice

Referee verdict: Incorrect sufficiency claim.

Assumptions, selection and testing gaps: Finite UV cutoff does not remove large finite Higgs corrections or establish naturalness.

Archive locator: <https://zenodo.org/records/19196942>.

Entry 66: No Supersymmetry

Paper/version: 15. **Audit depth:** S.

Indexed claim: 14 faces = SM, no room for SUSY

Referee verdict: Ontology hypothesis.

Assumptions, selection and testing gaps: A complete single-cell basis cannot exclude collective/multi-cell sectors; Kelvin premise overstated.

Archive locator: <https://zenodo.org/records/19064126>.

Entry 67: Baryon, Bekenstein, Cosmo

Paper/version: 49. **Audit depth:** S.

Indexed claim: η_B , $k = C_A = 3$, n-p mass diff

Referee verdict: Unestablished cosmology.

Assumptions, selection and testing gaps: Thermal transition and transport not derived; abundance ratio not a structure-formation theory; area quantum interpretation conditional.

Archive locator: <https://zenodo.org/records/19448089>.

Entry 68: Dark Energy 6/7 Factor**Paper/version:** 53. **Audit depth:** D.**Indexed claim:** $\rho - \Lambda$ from $(F - \chi)/F=6/7$, 0.47%**Referee verdict:** Identity plus unproved energy mapping.**Assumptions, selection and testing gaps:** 12/14 projector count valid; excluding stationary modes from energy and using observed radius are extra assumptions.**Archive locator:** <https://zenodo.org/records/19483955>.**Entry 69: Tensor-to-Scalar Ratio****Paper/version:** 55. **Audit depth:** D.**Indexed claim:** $r=0.0225$, testable by LiteBIRD**Referee verdict:** Revised hypothesis/arithmetic error.**Assumptions, selection and testing gaps:** Post-tension ratio-to-product switch; observed ns input; printed tensor tilt gives -0.0010 rather than -0.008.**Archive locator:** <https://zenodo.org/records/19484103>.**Entry 70: Baryon Asymmetry NLO****Paper/version:** 61. **Audit depth:** S.**Indexed claim:** $\eta = 6.109 \times 10^{-10}$, 0.09σ **Referee verdict:** Postdiction.**Assumptions, selection and testing gaps:** NLO baryogenesis correction selected; 951 rivals in tested grammar; transport and thermal source absent.**Archive locator:** <https://zenodo.org/records/19491332>.**Entry 71: Independent Convergence (Proton Radius, Vacuum Energy)****Paper/version:** 62. **Audit depth:** S.**Indexed claim:** $r_p = 4 \ell_P(m_P/m_p) = 0.8413 \text{ fm}$; independent convergence with Haramein 2012/2019**Referee verdict:** Postdiction/conditional.**Assumptions, selection and testing gaps:** Proton-radius factor and horizon-density mapping not derived; shared agreement is not independent physical confirmation.**Archive locator:** <https://zenodo.org/records/19624911>.**Entry 72: Graph-Fourier Companion to Paper #53 (6/7 Dark-Energy Factor)****Paper/version:** 70. **Audit depth:** D.**Indexed claim:** $\text{tr}(P_{\text{int}})/F = 6/7$ as propagating-mode fraction via band-edge zero-group-velocity lemma; Tier 1 algebraic identity**Referee verdict:** Confirmed projector/unsupported energy claim.

Assumptions, selection and testing gaps: Rank12 fraction reproduced; band-edge zero velocity is not zero energy or a vacuum stress tensor calculation.

Archive locator: <https://zenodo.org/records/19626516>.

Entry 73: Running Coupling Constants

Paper/version: 6. **Audit depth:** S.

Indexed claim: $b_0^{\text{EM}}=4/3$, $\alpha(M_Z)=1/128.95$

Referee verdict: Conditional/imported.

Assumptions, selection and testing gaps: Standard charged spectrum and RG calculations imported; hadronic contributions and thresholds need lineage.

Archive locator: <https://zenodo.org/records/19063473>.

Entry 74: Periodic Table (Aufbau)

Paper/version: 30. **Audit depth:** S.

Indexed claim: s,p,d,f from O_h crystal field

Referee verdict: Unestablished.

Assumptions, selection and testing gaps: Angular-momentum degeneracy imported; many-electron screening, shell ordering and exceptions not derived from face spectrum.

Archive locator: <https://zenodo.org/records/19185628>.

Entry 75: Visible Spectrum

Paper/version: 37. **Audit depth:** D.

Indexed claim: 7 irreps $\rightarrow$ 7 spectral bands

Referee verdict: Speculative/dimensional gap.

Assumptions, selection and testing gaps: Dimensionless 40.5 is assigned Hz; octave count/color/pineal correspondence has no derived scale or transduction.

Archive locator: <https://zenodo.org/records/19306316>.

Entry 76: Three Generations

Paper/version: 38. **Audit depth:** S.

Indexed claim: $N_{\text{gen}} = F_{\text{sq}}/2 = 3$

Referee verdict: Unproved identification.

Assumptions, selection and testing gaps: Three spatial axes identified as flavor; rotational invariance and distinct generation masses need proof.

Archive locator: <https://zenodo.org/records/19306393>.

Entry 77: Path Integral from Foam**Paper/version:** 42. **Audit depth:** S.**Indexed claim:** 5 Feynman rules from torsion walks**Referee verdict:** Conditional.**Assumptions, selection and testing gaps:** Walk sums legitimate; Wick rotation, spinors, interactions and QED vertices need a specified action.**Archive locator:** <https://zenodo.org/records/19306828>.**Entry 78: Information Capacity****Paper/version:** 43. **Audit depth:** S.**Indexed claim:** Bandwidth, Bekenstein $S \approx F=14$ **Referee verdict:** Unestablished.**Assumptions, selection and testing gaps:** Bit/channel capacities assumed; counting faces does not derive black-hole entropy or its normalization.**Archive locator:** <https://zenodo.org/records/19306918>.**Entry 79: Void Channel (v2.0)****Paper/version:** 45. **Audit depth:** D.**Indexed claim:** $H=L+\eta V$, boson-fermion parity; v2 §6 corrected per Paper #75 (η = formation rate, not correlation strength); supersedes [19307111](#)**Referee verdict:** Conditional.**Assumptions, selection and testing gaps:** Antipodal operator valid; parity is not statistics. Imprint/no-signalling and lab qubit premises remain open.**Archive locator:** <https://zenodo.org/records/21331511>.**Archive locator:** <https://zenodo.org/records/19307111>.**Entry 80: Spectral Verification****Paper/version:** 20. **Audit depth:** D.**Indexed claim:** Spectral Verification**Referee verdict:** Reproduction only.**Assumptions, selection and testing gaps:** Original script blocked by missing sympy; equivalent graph claims independently reproduced exactly.**Archive locator:** <https://zenodo.org/records/19079730>.**Entry 81: Spectral Uniqueness of the Truncated Octahedron Among Fedorov Parallelohedra****Paper/version:** 19625142. **Audit depth:** D.

Indexed claim: Pure mathematics (MSC 05C50, 52B10, 52C22). Independent combinatorial result; readable without UFFT context. Every claim reproduced by the bundled verification script.

Referee verdict: Restricted mathematics.

Assumptions, selection and testing gaps: Graph characterisation reproduced; eigenvalue7 square fraction wrong ($1/7$); hex decomposition omission; scope/optimalty overstated.

Archive locator: <https://zenodo.org/records/19625142>.

Index and source-access gaps

- Alpha paper #3 v2 and v3 Markdown files are named but absent; those versions were not independently inspected.
- No numbered #4 or #12 row is present in the index.
- The referenced `verification/peer_review_deliverables` directory is absent; its joint-statistics defense could not be certified.
- Repository text is not asserted identical to every archived Zenodo version.
- Earlier Bell and void-channel repository files are retained in the reproducibility package without an independent full-proof verdict.